LOUAY HAMMOUDI

Telecommunications & Networks Engineering Student | AI, Cybersecurity & Cloud

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PROFILE

Final-year ICT Engineering student at ENIT specializing in Converged Infrastructures, Cloud & Cybersecurity (I3C) and completing a concurrent IST Master's programme from September 2026. Hands-on experience in AI-driven security, SOC/WAF systems, RAG and agentic AI, cloud-native security, AWS, Docker, Kubernetes and Linux. Engineering projects cover C++20 robotic software-in-the-loop simulation and verification, privacy-aware access control, RF modelling and optimization, controlled network experimentation and reinforcement learning. Interested in research at the intersection of cybersecurity, AI/LLMs, networked systems and secure automation.

EDUCATION

2024-2027 — **National Engineering Degree in ICT - ENIT - National Engineering School of Tunis | Tunis**
Telecommunications & Networks. I3C specialization from September 2026; concurrent Master's programme in Information Systems Techniques (IST). Relevant coursework: Computer Networks, Routing, Network & System Administration, Mobile Networks, Digital Communications, Embedded Systems, Learning Algorithm Design for AI, RF Systems, Secure Web, Cloud & Network Security, IP/MPLS QoS, Virtualization & High Availability, Advanced Service Architectures and SDN; electives include Big Data, Data Mining, GeoAI & Geospatial Data, Semantic Web & Ontology, Artificial Intelligence, Advanced Image/Speech Processing, Pattern Recognition, Wireless Sensor Networks, System-on-Chip Design and Embedded Systems.

2022-2024 — **Preparatory Programme in Physics & Chemistry - IPEIM | Monastir**
National engineering-school entrance rank: 133/1750.

2021 — **Tunisian Baccalaureate in Experimental Sciences**
17.53/20, First Class Honours.

PROFESSIONAL EXPERIENCE

Dec 2025-May 2026 — **AI & Cybersecurity Intern - Sopra Steria | Tunis**
Developed NovaGuard AI for centralized monitoring and auditing of distributed ModSecurity WAF configurations. Combined WAF alerts, configuration data and technical documentation in RAG-based workflows to support security analysis and actionable recommendations.

Jun-Aug 2026 — **AI & Enterprise Software Intern - AYMAX | Tunis**
Built a VS Code multi-agent assistant for SAP Clean Core modernization and legacy-code analysis using RAG/LLM workflows, retrieval, tool-use patterns and agent orchestration. Connected outputs to ABAP RAP, CDS Views, OData services and Fiori Elements for enterprise development workflows.

Summer 2025 — **Digital Transformation Intern - SORECI | Tunis**
Supported network/system administration, infrastructure troubleshooting, optimization and AWS cloud-migration activities.

Apr-May 2025 — **Web Developer & AI Tools Database Engineer - Louati Construction (Italian Startup) | Remote**
Built a real-time project-tracking dashboard and integrated structured data for tasks, schedules and milestones; applied AI-assisted analysis to flag delivery risks and support assignment optimization.

2024-Present — **Web Developer - ENIT Junior Enterprise | Tunis**
Developed Next.js websites and custom tools for clients, including backend logic and APIs; translated non-technical requirements into working software.

Summer 2024 — **Data, AI & IoT Intern - Tagourti Car Company**
Designed a database for vehicle listings, customer records and sales data; developed a car recommendation tool and fair-price estimator, alongside hands-on IoT training.

SELECTED PROJECTS

2026 — **NovaGuard AI (Sopra Steria) - WAF Security Monitoring & RAG**
Developed centralized monitoring and auditing for distributed ModSecurity WAF configurations. Connected WAF alerts, configurations and documentation using RAG for security analysis and actionable recommendations.

2026 — **A-CorePilot (AYMAX) - Multi-Agent AI for Software Modernization**
Built a VS Code assistant with LLM/RAG workflows, retrieval, tool-use patterns and agent orchestration. Supported SAP Clean Core and legacy-code analysis with ABAP RAP, CDS Views, OData and Fiori Elements.

2026 — **MedVision - Safety-Aware Breast Histopathology AI & C++ Inference | GitHub**
Built a patient-disjoint ResNet-18 histopathology pipeline with clinical metrics, Grad-CAM, calibration and robustness checks. Exported ONNX models and implemented a C++17/OpenCV/ONNX Runtime API with latency benchmarking and deterministic safety reporting.

2026 — **RobSim V&V - C++20 Robotic Simulation & Software Verification | GitHub**
Migrated the deterministic software-in-the-loop robotic simulator from Python to C++20, modelling actuator dynamics, PID feedback control and virtual sensor responses. Designed configurable fault-injection scenarios, diagnostic and simulated safe-state handling, and reproducible verification metrics and reports.

Apr-May 2025 — **Intelligent Structural Analysis Platform (Louati Construction) | C++20, Eigen, Python, ML, RAG**
Developed a C++20 finite element analysis engine using the Direct Stiffness Method and Eigen for structural modelling, numerical computation and frame analysis. Built Python-based ML surrogate models for structural-response prediction and integrated a RAG-LLM assistant for technical-document retrieval, engineering explanations and automated reporting.

Summer 2024 — **Intelligent Vehicle Health & After-Sales Platform (Tagourti Car Company) | C++, Python, ML, RAG**
Developed a C++ diagnostic engine for vehicle-data processing, signal filtering, feature extraction and component health assessment. Implemented Python predictive-maintenance models and a RAG-LLM technical assistant for diagnostic interpretation, maintenance recommendations and report generation.

SELECTED PROJECTS (CONTINUED)

2026-Present **Edge SRE - Autonomous Offline AI Infrastructure Reliability & Recovery System | In Progress**
Designing a five-module offline-first reliability system: Monitoring Agent, AI Diagnostic Engine, Local LLM Agent, Secure Remediation Engine, and Verification & Observability for local incident detection and analysis. Planning policy-gated recovery, rollback safeguards, audit trails and post-remediation health checks. Architecture in development; autonomous remediation and recovery verification are not yet implemented.

Jun-Jul 2026 **Cloud-Native Security Monitor | GitHub**
Built a Kubernetes security-monitoring environment using Trivy, Falco, OPA and Prometheus/Grafana for scanning, policy enforcement and runtime visibility.

Jun-Jul 2026 **Network IDS & SOC Copilot | GitHub**
Built a Zeek/Suricata monitoring pipeline with ML attack-detection components, OpenSearch investigation and a RAG-assisted SOC copilot linked to MITRE ATT&CK.

Jun-Jul 2026 **Secure RAG Red-Team Benchmark | GitHub**
Designed tests for prompt injection, poisoned documents, sensitive-data leakage, unsafe tool use, multi-agent delegation issues and workflow-level attacks.

2026 **Network Impairment Harness | GitHub**
Built a reproducible Linux UDP test harness using network namespaces and tc (NetEm + TBF) to impose time-varying bandwidth, delay and packet loss. Compared imposed vs measured throughput, loss, one-way delay, p95 and reordering; investigated queueing, shaping accuracy and VM scheduling noise.

2026 **ARGUS-Micro - Reinforcement Learning for Sequential Cyber-Defence | GitHub**
Implemented a reproducible MDP environment with Dynamic Programming, Monte Carlo, TD(0), SARSA, Expected SARSA and Q-learning. Validated learned policies against exact value iteration across multi-seed experiments and built a Streamlit dashboard for policy/performance analysis.

2026 **AI & Cybersecurity Research Manuscript - Dr. Ines Bouzouiuta**
Research manuscript in preparation on the use of RAG and LLMs with ModSecurity data and security knowledge; investigating retrieval-augmented workflows for WAF analysis and security-rule engineering.

2024-2025 **Penetration-Testing Research Project - Dr. Hamza Hammami**
Designed a controlled lab with Kali Linux, Metasploit, Armitage and VirtualBox; defined reproducible attack scenarios and analyzed defensive hardening measures.

2024-2026 **Embedded AI, IoT & Applied Data Projects**
Built STM32/TensorFlow Lite prototypes for on-device inference and OpenCV/MQTT prototypes combining sensing, computer vision and IoT communication.

2026 **ECG Arrhythmia Classification - Time-Frequency + Deep Learning | Python, SciPy, PyWavelets, TensorFlow, MIT-BIH**
Built ECG preprocessing with band-pass filtering, baseline-wander removal and R-peak detection; converted beats to wavelet/spectrogram representations. Trained a CNN-LSTM for arrhythmia classification, handled class imbalance with SMOTE/weighted loss, and evaluated F1, sensitivity and specificity.

2026 **EHealthAccess - Privacy-Aware E-Health Access Control Prototype | GitHub**
Built a Node.js/Express policy engine for synthetic e-health data with stakeholder-based decisions, selective disclosure, pseudonymization and suppression. Developed a web dashboard, automated policy tests and demo audit logs; included reference Solidity and HLPSL artifacts for security-model exploration.

2026 **RF Exposure Modelling & PSO Optimization | GitHub**
Built a Python/NumPy model coupling antenna-array configuration, electromagnetic propagation, multilayer tissue interaction, SAR/thermal indicators, SINR, throughput, energy and cost. Formulated a constrained multi-objective Particle Swarm Optimization problem and added a Streamlit dashboard for analysis and parameter exploration.

2026 **Automatic Modulation Classification with Deep Learning | Python, PyTorch, NumPy/SciPy, RadioML**
Designed a CNN/ResNet model to classify 10+ digital modulation schemes from raw I/Q samples and evaluated performance from -20 to +18 dB SNR. Compared deep learning with cumulant + SVM features; achieved >90% accuracy at high SNR and analyzed confusion between similar modulations.

2026 **Jamming & Anomaly Detection in Wireless Signals | Python, scikit-learn, TensorFlow/Keras**
Simulated AWGN, fading and narrowband/wideband/sweep jamming; extracted STFT/spectrogram features for interference analysis. Trained an unsupervised autoencoder and CNN classifier for real-time anomaly/jamming detection with low false-positive behavior.

2026 **Modern C++ Algorithms & Data Structures | GitHub**
Implemented a header-only C++20 library with AVL trees, separate-chaining hash tables, binary heaps, sorting/search algorithms, BFS/DFS, Dijkstra shortest paths and topological sorting. Applied C++20 concepts and ranges, STL and smart-pointer ownership; added CMake-based tests and documented time/space complexity for the algorithms and data structures.

TECHNICAL SKILLS

Programming	Python, C++20, C, C#, Java, JavaScript, Embedded C, SQL, Node.js, Express, Solidity
C++ / Algorithms	C++20 concepts/ranges, templates, STL, CMake, RAII / smart pointers, AVL trees, hashing, heaps, graph algorithms
AI / GenAI	ML/DL, NLP, RAG, agent orchestration, prompt chains, LLM evaluation/red teaming, TensorFlow/Keras, PyTorch, scikit-learn, Hugging Face, LangChain, LangSmith, LlamaIndex, MCP, AutoGen, CrewAI, Langflow, Flowise, embeddings, vector databases, Grad-CAM, ONNX/ONNX Runtime, OpenCV
Software V&V / Robotics	C++20 robotic simulation, software-in-the-loop (SiL), PID control, actuator/sensor modelling, discrete-time simulation, fault injection, Pytest (Python baseline), automated verification, metrics and reports, YAML scenarios, GitHub Actions CI
Security / Privacy	Role- and context-based access control, data masking, pseudonymization, consent-aware policy prototypes, audit event logging, Solidity reference contracts, HLPSL security modelling
SRE / Reliability	SRE concepts, observability, monitoring, incident triage, fault injection, offline-first edge architecture, policy-gated remediation design, rollback planning and post-recovery verification (ongoing)
Scientific / RL	Numerical modelling, multi-objective optimization, Particle Swarm Optimization, reinforcement learning, MDPs, Dynamic Programming, Monte Carlo, TD learning, SARSA, Q-learning, experimental validation, time-series analysis
RF / Signals	RF systems, electromagnetic propagation, antenna arrays, SINR, link-budget reasoning, signal processing, I/Q analysis, STFT/spectrograms, advanced cellular networks, wireless sensor networks, optical-network fundamentals
Cyber / Networks	TCP/IP, routing, Linux administration, SOC, WAF, SIEM, ModSecurity, Zeek, Suricata, MITRE ATT&CK, penetration testing, MQTT, Linux network namespaces, tc/NetEm/TBF, IP/MPLS QoS, SDN
Cloud / DevSecOps	AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, OpenSearch, Trivy, Falco, OPA, Prometheus, Grafana, Git, virtualization, high availability, cloud monitoring
SAP / Enterprise	SAP Clean Core, ABAP RAP, CDS Views, OData, Fiori Elements, legacy-code analysis, governance checks
Embedded / Data	STM32, RTOS, Arduino, Raspberry Pi, PostgreSQL, MySQL, MongoDB, Streamlit, NumPy, SciPy, PyWavelets, Matplotlib

CERTIFICATIONS, LANGUAGES & ACTIVITIES

Certifications & training: Cisco CCNA | AWS Academy Cloud Foundations, Developing and Operations | Linux Administration | CTF Architecture

Languages: Arabic (Native) | French (C1) | English (C1) | German (A2)

Activities: Two-time Tunisian national taekwondo champion and former national-team member; Sécurinets Club CTF organization and Linux/exploit-development training.